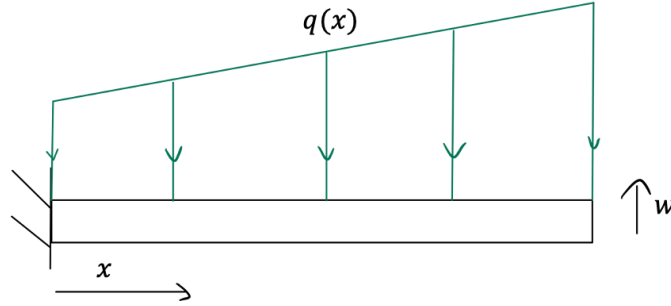


Instructions:

- Submit your assignment as a single PDF file through Canvas.
- Show all your steps and solution procedures.
- Make sure that your solution is well organised and that you are using appropriate headers for each question and sub-question.
- Scanned photos of your handwritten solution are acceptable as long as they are **legible**
- If a question requires plotting, please plot the expressions using any software/language (MATLAB, Excel, Mathematica, etc.) of your choice. However, **hand drawings** will not be accepted.
- You can use any software or calculator to perform any calculations. Using symbolic packages in MATLAB, Mathematica is acceptable as long as the solution steps are well laid out in your submission. In case you use those packages please attach the script in your submission file.
- While you can discuss the problems with classmates, all submitted work must be your own to conform to academic integrity guidelines.
- Late assignments will not be accepted.

Question 1 (50 Points)

The transverse deflection $w(x)$ of a cantilever beam due to a distributed load $q(x)$ can be modelled using EQ 1 below. The weak form of **EQ 1** is shown in **EQ 2** where $v(x)$ represents the weight function. **EQ 3** and **EQ 4** show a possible approximate solution $W(x)$ and weight function using the Ritz/Galerkin method.



$$EI \frac{d^4 w}{dx^4} = q(x) \quad 0 \leq x \leq L \quad (\text{EQ 1})$$

where $q(x) = -950x - 50$, $L = 1 \text{ m}$ and $EI = 10^6 \text{ Nm}^2$

$$EI \int_0^L \frac{d^2 v}{dx^2} \frac{d^2 w}{dx^2} dx - \left(\frac{dv}{dx} EI \frac{d^2 w}{dx^2} \right) \Big|_0^L + \left(v EI \frac{d^3 w}{dx^3} \right) \Big|_0^L = \int_0^L v q dx \quad (\text{EQ 2})$$

$$w(x) \approx W(x) = c_2 x^3 + c_1 x^2 \quad (\text{EQ 3})$$

$$v(x) = x^3 + x^2 \quad (\text{EQ 4})$$

$$i. e. \theta_1 = x^2 \text{ \& } \theta_2 = x^3$$

- Show that the approximate solution satisfies the essential boundary conditions at the fixed end ($x = 0$).
- Solve for the coefficients c_1 and c_2
- Compare the approximate solution $W(x)$ to the exact solution $w(x)$ (**EQ 5**) obtained in assignment 1 by plotting $w(x)$ and $W(x)$ on the same graph.

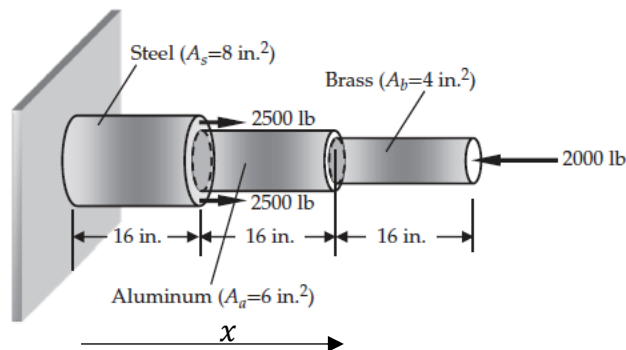
$$w(x) = \frac{1}{EI} \left(-\frac{950}{120} x^5 - \frac{50}{24} x^4 + \frac{525}{6} x^3 - \frac{1025}{6} x^2 \right) \quad (\text{EQ 5})$$

In this assignment, you can directly use the following formulas to define the stiffness matrix and force vectors for axial bars. There is no need to derive them from the governing equation.

Linear Element	
$[k_{ij}^e] = EA \int_{x_1^e}^{x_2^e} \frac{d\psi_i^e}{dx} \frac{d\psi_j^e}{dx} dx$	$\{f_i\} = \int_{x_1^e}^{x_2^e} \psi_i^e b(x) dx$
$\psi_1^e = \frac{x_2^e - x}{L_e}$	$\psi_2^e = \frac{x - x_1^e}{L_e}$

Question 2 (50 Points)

For the stepped bar shown below, use **3 linear** 1D elements to model the axial displacement in the bar. You can assume that there are no body forces acting on the system.



- I. Evaluate the element wise stiffness matrices and force vectors.
- II. Assemble the matrices into a global system.
- III. Apply the boundary conditions and solve for the nodal displacements.
- IV. Plot the displacement, strain, and stress as a function of the axial position x .

E_{steel}	$30 * 10^6 \text{ psi}$
$E_{aluminum}$	$10 * 10^6 \text{ psi}$
E_{brass}	$15 * 10^6 \text{ psi}$

NB: To facilitate the grading please assign elements 1,2 and 3 to the steel, aluminum and brass sections respectively.